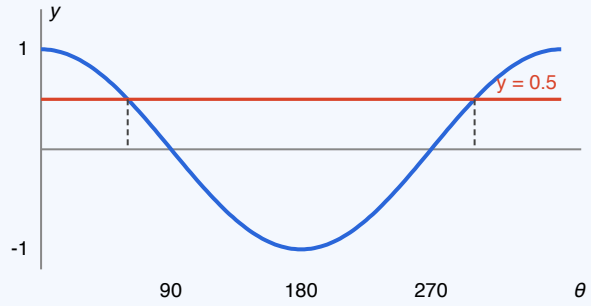
Shape and Space 10 — Graphs of sin, cos and tan · Sine rule · Cosine rule · Area of a triangle

Calculator allowed. Marks in brackets. ★ = harder question. Lengths to 3 s.f., angles to 1 d.p. Diagrams not drawn accurately.

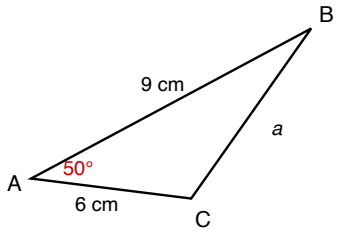
1 **a** Sketch the graph of $y = \cos\theta$ for $0° \le \theta \le 360°$. **b** Use your graph to find all solutions of $\cos\theta = \dfrac{1}{2}$ in that range. [7]

a Starts at 1, falls through 0 at $90°$ to -1 at $180°$, back through 0 at $270°$ to 1 at $360°$.



b The line $y = \frac{1}{2}$ meets the curve twice: $\theta = 60°$ (from the calculator) and, by the symmetry of the cosine graph, $\theta = 360° - 60° = 300°$.

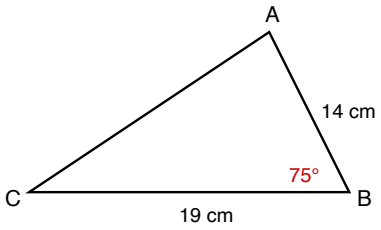
2 In triangle ABC, angle A $= 50°$, AB $= 9$ cm and AC $= 6$ cm. Find **a** the length a = BC **b** angle B. [6]



a Two sides and the included angle → cosine rule:
$a^2 = 9^2 + 6^2 - 2 \times 9 \times 6\cos 50° = 47.58$, so $a = 6.90$ cm.

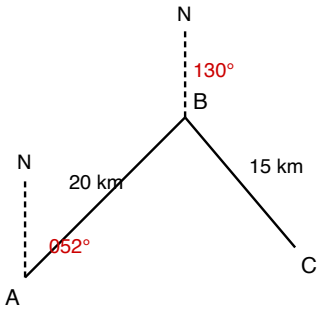
b Now a side and its opposite angle are known → sine rule:
$$\frac{\sin B}{6} = \frac{\sin 50°}{6.90} \Rightarrow \sin B = 0.6663 \Rightarrow B = 41.8°.$$

3 In triangle ABC, angle B $= 75°$, AB $= 14$ cm and BC $= 19$ cm. Find **a** angle C **b** the area of the triangle. [6]



a First AC by the cosine rule: $b^2 = 14^2 + 19^2 - 2 \times 14 \times 19 \cos 75° = 419.3$, $b = 20.48$ cm. Then the sine rule: $\dfrac{\sin C}{14} = \dfrac{\sin 75°}{20.48} \Rightarrow \sin C = 0.6604 \Rightarrow C = 41.3°$.
b Area $= \frac{1}{2} \times 14 \times 19 \times \sin 75° = 128.5$ cm² (half the product of two sides and the sine of the angle between them).

4 Town B is 20 km from town A on a bearing of 052°. Town C is 15 km from B on a bearing of 130°. **a** Calculate the direct distance AC. **b** Calculate the bearing of C from A. [6]



a The angle at B inside the triangle: the bearing of A from B is $052° + 180° = 232°$, so angle ABC $= 232° - 130° = 102°$. Cosine rule:
$\mathrm{AC}^2 = 20^2 + 15^2 - 2 \times 20 \times 15 \cos 102° = 749.7$, $\mathrm{AC} = 27.4$ km.
b Sine rule for angle BAC: $\dfrac{\sin A}{15} = \dfrac{\sin 102°}{27.38} \Rightarrow A = 32.4°$. Bearing of C from A $= 052° + 32.4° = 84°$ (3 figures: $084°$).

Total: 25 marks

Check: sine graph starts at 0, cosine at 1; both repeat every 360° · second solution: sin → $180° - \theta$, cos → $360° - \theta$ · two sides + included angle, or three sides → cosine rule; otherwise sine rule · area $= \frac{1}{2}ab\sin C$ · bearings: find the angle inside the triangle first, using north lines and $180°$ back-bearings.